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Inventor(s) OGASAWARA; Yoshiyasu et al.

ELECTRONIC DEVICE

Abstract

An electronic device includes a first device and a second device. The first device and the second device are removably attachable to each other. The first device includes a first front surface, and a first display and a first connection portion arranged at the first front surface. The second device includes a second front surface, a second display arranged at the second front surface, a second rear surface located opposite to the second front surface, and a second connection portion arranged at the second rear surface. The second connection portion is connectable to the first connection portion in a first direction and a second direction opposite to the first direction. In a first connection state in which the second connection portion is connected to the first connection portion in the first direction, the second rear surface is arranged to cover the first display.

Inventors: **OGASAWARA; Yoshiyasu (Kyoto-shi, JP), TSUCHIYA; Hitoshi (Kyoto-shi, JP), NIWA; Masato (Kyoto-shi, JP), SATO; Takahiro (Kyoto-shi, JP), TANIGUCHI; Nonoho (Kyoto-shi, JP)**

Applicant: **Nintendo Co., Ltd.** (Kyoto-shi, JP)

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Background/Summary

[0001] This nonprovisional application claims priority on and is a continuation of International Patent Application PCT/JP2022/042086 filed with the Japan Patent Office on Nov. 11, 2022, the entire contents of which are hereby incorporated by reference.

FIELD

[0002] The present disclosure relates to an electronic device.

BACKGROUND AND SUMMARY

[0003] A dual display mobile device has been known.

[0004] An exemplary embodiment provides an electronic device that includes a first device and a second device. The first device and the second device are removably attachable to each other. The first device includes a first front surface, and a first display and a first connection portion arranged at the first front surface. The second device includes a second front surface, a second display arranged at the second front surface, a second rear surface located opposite to the second front surface, and a second connection portion arranged at the second rear surface. The second connection portion is connectable to the first connection portion in a first direction and a second direction opposite to the first direction. In a first connection state in which the second connection portion is connected to the first connection portion in the first direction, the second rear surface is arranged to cover the first display. In a second connection state in which the second connection portion is connected to the first connection portion in the second direction, the second rear surface is arranged not to cover the first display. Thus, a novel structure that switches between connection states is provided.

[0005] For example, in communication between the first device and the second device, an external terminal may be provided in each of the first device and the second device and the external terminals may be connected to each other. When the external terminal is provided in each of the first device and the second device, however, high-frequency parasitic emission may occur.

[0006] According to the electronic device according to the above, the first device may include a first substrate arranged inside the first device and one first antenna or at least two first antennas provided at the first substrate. The second device may include a second substrate arranged inside the second device and one second antenna or at least two second antennas provided at the second substrate. In each of the first connection state and the second connection state, when the first display is viewed in a plan view, the first antenna and the second antenna may be arranged at positions superimposed on each other. The first device and the second device may be configured to perform first wireless communication through electromagnetic field coupling between the first antenna and the second antenna. Thus, without providing the external terminal, in each of the first connection state and the second connection state, the first device and the second device can communicate with each other. Therefore, occurrence of high-frequency parasitic emission can be suppressed. Furthermore, necessity for measures against electrostatic discharge can be reduced.

[0007] According to the electronic device according to the above, in each of the first connection state and the second connection state, at least a part of the first antenna may be arranged at a position superimposed on the second antenna. Strength of electromagnetic field coupling can thus be enhanced.

[0008] According to the electronic device according to the above, one of the first antenna or the second antenna may include a first antenna portion. The other of the first antenna or the second antenna may include a second antenna portion and a third antenna portion. In the first connection state, the first antenna portion may be arranged as being superimposed on the second antenna portion. In the second connection state, the first antenna portion may be arranged as being superimposed on the third antenna portion. The number of antenna portions can thus be set to three. The number of antenna portions can be smaller than in an example where two antenna portions are arranged in the first device and two antenna portions are arranged in the second device.

[0009] According to the electronic device according to the above, the first antenna may include the first antenna portion. The second antenna may include the second antenna portion and the third antenna portion. When the first front surface is viewed in the plan view, the first antenna portion may be arranged at a position adjacent to the first connection portion. When the second rear surface is viewed in the plan view, the second connection portion may be arranged as lying between the second antenna portion and the third antenna portion.

[0010] According to the electronic device according to the above, the first device may include a first substrate arranged inside the first device and one first antenna or at least two first antennas provided at the first substrate. The second device may include a second substrate arranged inside the second device and one second antenna or at least two second antennas provided at the second substrate. One of the first antenna or the second antenna may include a first antenna portion. The other of the first antenna or the second antenna may include a second antenna portion and a third antenna portion. In the first connection state, the second antenna portion may be arranged as being superimposed on the first antenna portion and the third antenna portion may be arranged as not being superimposed on the first antenna. In the second connection state, the third antenna portion may be arranged as being superimposed on the first antenna portion and the second antenna portion may be arranged as not being superimposed on the first antenna.

[0011] According to the electronic device according to the above, when the first front surface is viewed in a plan view, the first antenna portion may be arranged at a position adjacent to the first connection portion. When the second rear surface is viewed in the plan view, the second connection portion may be arranged as lying between the second antenna portion and the third antenna portion.

[0012] According to the electronic device according to the above, in the second connection state, the first device and the second device may be connected with the second display being inclined with respect to the first display.

[0013] According to the electronic device according to the above, in the first connection state, the first device and the second device may be connected with the second display being in parallel to the first display.

[0014] According to the electronic device according to the above, the first connection portion may include a first inclined surface inclined with respect to the first front surface. The second connection portion may include a second inclined surface inclined with respect to the second rear surface.

[0015] According to the electronic device according to the above, an angle between the first antenna and the second antenna in a cross-section perpendicular to each of the first front surface and the second rear surface in the first connection state is defined as a first angle. An angle between the first antenna and the second antenna in the cross-section perpendicular to each of the first front surface and the second rear surface in the second connection state is defined as a second angle. An angle between the first front surface and the first inclined surface in the cross-section perpendicular to the first front surface is defined as a third angle. An angle between the second rear surface and the second inclined surface in the cross-section perpendicular to the second rear surface is defined as a fourth angle. Each of the first angle and the second angle may be smaller than a total of the third angle and the fourth angle. Thus, in each of the first connection state and the second connection state, high strength of electromagnetic coupling can be maintained.

[0016] According to the electronic device according to the above, when the first antenna and the second antenna are electromagnetically coupled to each other, the first device and the second device may transmit and receive a signal unidirectionally or bidirectionally. The first connection portion may include a first contact for electric power supply. The second connection portion may include a second contact for electric power supply. In each of the first connection state and the second connection state, the first contact and the second contact may be connected. Electric power can thus be supplied with the use of the first contact and the second contact. Since a terminal for electric power supply is less likely to be affected by parasitic emission, wired connection can be established.

[0017] According to the electronic device according to the above, the first device and the second device may be configured to perform second wireless communication while the first device and the second device are separate. Thus, even while the first device and the second device are separate, the first device and the second device can communicate with each other.

[0018] According to the electronic device according to the above, the first wireless communication in the first connection state and the second connection state may be performed through electromagnetic field coupling between the first antenna and the second antenna. The second wireless communication while the first device and the second device are separate may be performed through electromagnetic field coupling between antennas different from each of the first antenna and the second antenna. A frequency band of a signal to be used in the first wireless communication may be higher than a frequency band of a signal to be used in the second wireless communication. In general, as the frequency is higher, higher directivity is demanded. By setting the frequency band of the signal to be used in second wireless communication to be lower than the frequency band of the signal to be used in first wireless communication, communication between the first device and the second device can be ensured even when the first device is distant from the second device.

[0019] According to the electronic device according to the above, the first connection portion may include a first receiving groove and a second receiving groove located opposite to the first receiving groove. The second connection portion may include a movable hook and a fixed hook located opposite to the movable hook. In the first connection state, the movable hook may be not arranged in the first receiving groove and the fixed hook may be arranged in the second receiving groove. In the second connection state, the movable hook may be arranged in the second receiving groove and the fixed hook may be arranged in the first receiving groove. Thus, in the first connection state, the second device can easily be removed from the first device. In the second connection state, on the other hand, the second device can securely be fixed to the first device.

[0020] According to the electronic device according to the above, the first connection portion may include a first electrical connection portion and a first mechanical connection portion. The first electrical connection portion may be provided above or below the first display. The first mechanical connection portion may be provided to sandwich the first electrical connection portion from a longitudinal direction of the first display. The second connection portion may include a second electrical connection portion and a second mechanical connection portion. The second electrical connection portion may be provided above or below the second display. The second mechanical connection portion may be provided to sandwich the second electrical connection portion from a longitudinal direction of the second display. The first electrical connection portion may electrically be connected to the second electrical connection portion. The first mechanical connection portion may mechanically be connected to the second mechanical connection portion.

[0021] When the mechanical connection portion is provided to sandwich the electrical connection portion from a direction of a short side of the display, an additional space in the direction of the short side is required. On the other hand, there is an extra space at each of opposing sides of the electrical connection portion in the longitudinal direction of the display. As the mechanical connection portion is provided to sandwich the electrical connection portion from the longitudinal direction of the display, the space can effectively be made use of.

[0022] According to the electronic device according to the above, the first electrical connection portion may have terminals disposed in the longitudinal direction of the first display. The second electrical connection portion may have a terminal disposed in the longitudinal direction of the second display. In an example where the terminal of the electrical connection portion is disposed in the longitudinal direction of the display, a length of the display in the direction of the short side can be longer than in an example where the terminal of the electrical connection portion is disposed in the direction of the short side of the display. Consequently, an area of the display can be larger.
[0023] According to the electronic device according to the above, the first device may include an input portion configured to be operated as a stick or a button. The second display may include a touch panel on which a touch input can be provided. Since the first device includes the input portion configured to be operated as the stick or the button, an operation suitable for an application can be performed. In the first connection state, the input portion configured to be operated as the stick or the button is covered with the second device. Therefore, when the electronic device is carried, the input portion is less likely to be caught by something else.
[0024] According to the electronic device according to the above, the first device may include a first housing that surrounds the first display and includes a first projecting portion that extends in an outward direction of the first display. The second device may include a second housing that surrounds the second display and includes a second projecting portion that extends in an outward direction of the second display. The first connection portion may be provided at a front side of the first projecting portion. The second connection portion may be provided at a rear side of the second projecting portion. Thus, the first device and the second device can be connected to each other while the first display and the second display can be increased in size.
[0025] According to the electronic device according to the above, the first device may include a first rear surface, a first upper surface, a first lower surface, a first right side surface, and a first left side surface. The first rear surface may be located opposite to the first front surface. The first upper surface may be contiguous to the first front surface and the first rear surface. The first lower surface may be located opposite to the first upper surface. The first right side surface may be contiguous to each of the first front surface, the first rear surface, the first upper surface, and the first lower surface. The first left side surface may be located opposite to the first right side surface. The second device may include a second upper surface, a second lower surface, a second right side surface, and a second left side surface. The second upper surface may be contiguous to the second front surface and the second rear surface. The second lower surface may be located opposite to the second upper surface. The second right side surface may be contiguous to each of the second front surface, the second rear surface, the second upper surface, and the second lower surface. The second left side surface may be located opposite to the second right side surface. The first projecting portion may be provided at the first upper surface. The second projecting portion may be provided at the second upper surface. The first device may include a first input portion provided at a portion of the first projecting portion at a side of the first right side surface and a second input portion provided at the first upper surface other than the first projecting portion. The second device may include a third input portion provided at a portion of the second projecting portion at a side of the second right side surface and a fourth input portion provided at the second upper surface other than the second projecting portion.
[0026] Thus, a mix-up between the first input portion and the second input portion can be less likely. Similarly, a mix-up between the third input portion and the fourth input portion can be less likely. In the second connection state, the fourth input portion of the second device is arranged at a position distant from the first device. Therefore, interference with operation of the first input portion is less likely. While the first device and the second device are separate, the input portions are provided at positions in common, and hence the first device and the second device can be operated with similar feeling. Furthermore, by horizontally sliding a finger, two different inputs can be provided.

[0027] According to the electronic device according to the above, in the first connection state, in a direction perpendicular to the first front surface, the first input portion and the third input portion may be arranged as being aligned and the second input portion and the fourth input portion may be arranged as being aligned. A user can thus more readily learn positions of the input portions.
[0028] According to the electronic device according to the above, at a surface of the second projecting portion, an input portion or a camera lens may be provided. A functional component that can be used in each of the first connection state and the second connection state can be obtained. Therefore, while the second projecting portion is used as the connection portion, a region in the second device other than the second projecting portion can effectively be made use of. For example, the second display can be increased in size.
[0029] According to the electronic device according to the above, the second display may include a touch panel on which a touch operation can be performed and a touch panel cover on which an input operation to the touch panel can be performed, the touch panel cover covering the touch panel. The second display may include a first tactile feel presentation portion configured with the touch panel cover and a second tactile feel presentation portion where the touch panel is exposed in an opening provided in the touch panel cover. Thus, when the second display includes the touch panel, the user can readily know the position of the second tactile feel presentation portion.
[0030] The foregoing and other objects, features, aspects, and advantages of the present disclosure will become more apparent from the following detailed description of the present disclosure when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0031] FIG. **1** shows an exemplary illustrative non-limiting drawing of a schematic perspective view showing a configuration of an electronic device according to a first embodiment.
[0032] FIG. **2** shows an exemplary illustrative non-limiting drawing of a schematic front view showing a configuration of a first device.
[0033] FIG. **3** shows an exemplary illustrative non-limiting drawing of an enlarged schematic diagram of a region III in FIG. **2**.
[0034] FIG. **4** shows an exemplary illustrative non-limiting drawing of a schematic front view showing a configuration of a second device.
[0035] FIG. **5** shows an exemplary illustrative non-limiting drawing of a partially enlarged view of a second rear surface of the second device.
[0036] FIG. **6** shows an exemplary illustrative non-limiting drawing of a schematic front view showing a configuration of the electronic device in a first connection state.
[0037] FIG. **7** shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view along the line VII-VII in FIG. **6**.
[0038] FIG. **8** shows an exemplary illustrative non-limiting drawing of a schematic diagram showing relation between a first connection portion and a second connection portion in the first connection state.
[0039] FIG. **9** shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view along the line IX-IX in FIG. **6**.
[0040] FIG. **10** shows an exemplary illustrative non-limiting drawing of a schematic front view showing a configuration of the electronic device in a second connection state.
[0041] FIG. **11** shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view along the line XI-XI in FIG. **10**.
[0042] FIG. **12** shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view showing a state in which the electronic device in the second connection state is

unlocked.
[0043] FIG. **13** shows an exemplary illustrative non-limiting drawing of a schematic diagram showing relation between the first connection portion and the second connection portion in the second connection state.
[0044] FIG. **14** shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view showing positional relation between a first antenna and a second antenna in the first connection state.
[0045] FIG. **15** shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view showing positional relation between the first antenna and the second antenna in the second connection state.
[0046] FIG. **16** shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view showing a modification of positional relation between the first antenna and the second antenna in the first connection state.
[0047] FIG. **17** shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view showing a modification of positional relation between the first antenna and the second antenna in the second connection state.
[0048] FIG. **18** shows an exemplary illustrative non-limiting drawing of a partial schematic perspective view showing a configuration of the electronic device according to a second embodiment.
[0049] FIG. **19** shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view along the line XIX-XIX in FIG. **18**.
[0050] FIG. **20** shows an exemplary illustrative non-limiting drawing of a schematic front view showing a configuration of the first device of the electronic device according to a third embodiment.
[0051] FIG. **21** shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view showing a configuration of a fifth input portion and a sixth input portion.
[0052] FIG. **22** shows an exemplary illustrative non-limiting drawing of a schematic front view showing a configuration of the second device of the electronic device according to the third embodiment.
[0053] FIG. **23** shows an exemplary illustrative non-limiting drawing of a schematic cross-sectional view along the line XXIII-XXIII in FIG. **22**.
[0054] FIG. **24** shows an exemplary illustrative non-limiting drawing of a schematic perspective view showing a configuration of the electronic device according to the third embodiment in the first connection state.
[0055] FIG. **25** shows an exemplary illustrative non-limiting drawing of a schematic perspective view showing a configuration of the electronic device according to the third embodiment in the second connection state.

DETAILED DESCRIPTION OF NON-LIMITING EXAMPLE EMBODIMENTS

[0056] An embodiment of the present disclosure will be described in detail with reference to the drawings. The same or corresponding elements in the drawings have the same reference characters allotted and description thereof will not be repeated.

First Embodiment

[0057] A configuration of an electronic device **300** according to a first embodiment will initially be described.
[0058] FIG. **1** is a schematic perspective view showing the configuration of electronic device **300** according to the first embodiment. As shown in FIG. **1**, electronic device **300** according to the first embodiment includes a first device **100** and a second device **200**. First device **100** and second device **200** are removably attachable to each other. Second device **200** is attachable to first device **100** and removable from first device **100**.
[0059] As shown in FIG. **1**, first device **100** includes a first front surface **101** and a first rear surface

102. First rear surface **102** is located opposite to first front surface **101**. Second device **200** includes a second front surface **201**, a second rear surface **202**, and a second display **210**. Second rear surface **202** is located opposite to second front surface **201**. Second display **210** is arranged at second front surface **201**. Second display **210** may be a touch panel on which a touch input can be provided.

[0060] FIG. **2** is a schematic front view showing a configuration of first device **100**. First device **100** includes a first upper surface **103**, a first lower surface **104**, a first right side surface **105**, and a first left side surface **106**. First upper surface **103** is contiguous to each of first front surface **101**, first rear surface **102**, first left side surface **106**, and first right side surface **105**. First lower surface **104** is located opposite to first upper surface **103**. First right side surface **105** is contiguous to each of first front surface **101**, first rear surface **102**, first upper surface **103**, and first lower surface **104**. First left side surface **106** is located opposite to first right side surface **105**. First left side surface **106** is contiguous to each of first front surface **101**, first rear surface **102**, first upper surface **103**, and first lower surface **104**.

[0061] As shown in FIG. **2**, first device **100** may further include a first housing **165**, a first display **110**, a first connection portion **120**, a ninth input portion **121**, an operation button **190**, an unlocking button **191**, and a Hall element **130**. First display **110** is arranged at first front surface **101**. First display **110** is attached to first housing **165**.

[0062] First connection portion **120** is arranged at first front surface **101**. First connection portion **120** is attached to first housing **165**. First connection portion **120** may project from a surface of first housing **165**. When first display **110** is viewed in a plan view, first connection portion **120** is arranged between first upper surface **103** and first display **110**. First display **110** being viewed in the plan view corresponds to first device **100** being viewed in a direction from first front surface **101** toward first rear surface **102**.

[0063] As shown in FIG. **2**, ninth input portion **121** is arranged, for example, at first front surface **101**. Ninth input portion **121** can be operated as a stick or a button. Unlocking button **191** is arranged, for example, at first upper surface **103**. Operation button **190** is arranged, for example, at first upper surface **103**. Hall element **130** is arranged inside first housing **165**. First device **100** does not have to include ninth input portion **121**.

[0064] FIG. **3** is an enlarged schematic diagram of a region III in FIG. **2**. As shown in FIG. **3**, first connection portion **120** includes, for example, a first mechanical connection portion **180**, a first electrical connection portion **160**, and a first magnetic connection portion **111**. First mechanical connection portion **180** includes a first main body portion **183** and a pair of first protruding portions **184**. First main body portion **183** is attached to first housing **165**. The pair of first protruding portions **184** is provided at first main body portion **183**. First main body portion **183** is provided with a first opening portion **185**.

[0065] First electrical connection portion **160** is exposed through first opening portion **185**. First electrical connection portion **160** includes a first contact **162** and a first holding portion **161**. First contact **162** is conductive. First holding portion **161** is insulating. First contact **162** is held by first holding portion **161**. Though the number of first contacts **162** is not particularly limited, for example, nine first contacts are provided. When first display **110** is viewed in the plan view, first contacts **162** are arranged along a line. One of first contacts **162** is, for example, a terminal for electric power supply.

[0066] First magnetic connection portion **111** is configured, for example, with a pair of magnets. First magnetic connection portion **111** is located between first main body portion **183** and first rear surface **102**. First magnetic connection portion **111** is arranged inside first device **100**. As shown in FIG. **3**, when first front surface **101** is viewed in the plan view, first electrical connection portion **160** is located as lying between the pair of magnets. In the present embodiment, when first front surface **101** is viewed in the plan view, first electrical connection portion **160** is located as lying between the pair of magnets in a longitudinal direction of first display **110** of first device **100**.

[0067] First device **100** includes a first substrate **131** and a first antenna **132**. Each of first substrate **131** and first antenna **132** is arranged inside first device **100**. More particularly, each of first substrate **131** and first antenna **132** is arranged inside first housing **165**. First antenna **132** is provided at first substrate **131**.

[0068] The number of first antennas **132** is not particularly limited. The number of first antennas **132** is set, for example, to one or at least two. In electronic device **300** according to the present embodiment, a single first antenna **132** is provided. First antenna **132** is, for example, a first antenna portion **132**. When first front surface **101** is viewed in the plan view, first antenna portion **132** may be arranged at a position adjacent to first connection portion **120**. First antenna portion **132** is arranged, for example, between first connection portion **120** and first right side surface **105**.

[0069] As shown in FIG. **3**, when first front surface **101** is viewed in the plan view, first antenna **132** may be located on an outer side of first connection portion **120**. From another point of view, when first front surface **101** is viewed in the plan view, first antenna **132** may be arranged at a position not superimposed on first main body portion **183**. For example, resin is employed as a material for first housing **165**. Though the material for first main body portion **183** is not particularly limited, for example, metal may be adopted. First antenna **132** may be arranged such that first main body portion **183** does not cut off electromagnetic field.

[0070] FIG. **4** is a schematic front view showing a configuration of second device **200**. As shown in FIG. **4**, second device **200** further includes a second upper surface **203**, a second lower surface **204**, a second right side surface **205**, and a second left side surface **206**. Second upper surface **203** is contiguous to second front surface **201** and second rear surface **202**. Second lower surface **204** is located opposite to second upper surface **203**. Second right side surface **205** is contiguous to each of second front surface **201**, second rear surface **202**, second upper surface **203**, and second lower surface **204**. Second left side surface **206** is located opposite to second right side surface **205**. Second left side surface **206** is contiguous to each of second front surface **201**, second rear surface **202**, second upper surface **203**, and second lower surface **204**.

[0071] As shown in FIG. **4**, second device **200** includes a second housing **265** and a magnet portion **230**. Second display **210** is attached to second housing **265**. Magnet portion **230** is arranged inside second housing **265**. As shown in FIG. **4**, when second display **210** is viewed in the plan view, magnet portion **230** may be located, for example, between second display **210** and second right side surface **205**. Second display **210** being viewed in the plan view corresponds to second device **200** being viewed in a direction from second front surface **201** toward second rear surface **202**.

[0072] FIG. **5** is a partially enlarged view of second rear surface **202** of second device **200**. As shown in FIG. **5**, second device **200** includes a second connection portion **220**. Second connection portion **220** is arranged at second rear surface **202**. Second connection portion **220** includes, for example, a second mechanical connection portion **280**, a second electrical connection portion **260**, and a second magnetic connection portion **211**. Second mechanical connection portion **280** includes a second main body portion **283**, a fixed hook **281**, and a movable hook **270**. Second main body portion **283** is attached to second housing **265**. Fixed hook **281** is fixed to second main body portion **283**. Movable hook **270** is attached to second main body portion **283** as being movable.

[0073] Second main body portion **283** is provided with a second opening portion **285** and a third opening portion **284**. Movable hook **270** is exposed through third opening portion **284**. Movable hook **270** is located opposite to fixed hook **281**. When second rear surface **202** is viewed in the plan view, second electrical connection portion **260** is located between movable hook **270** and fixed hook **281**. Second electrical connection portion **260** is exposed through second opening portion **285**. When second rear surface **202** is viewed in the plan view, fixed hook **281** is located between second upper surface **203** and movable hook **270**.

[0074] Second electrical connection portion **260** includes a second contact **262** and a second holding portion **261**. Second contact **262** is conductive. Second holding portion **261** is insulating. Second contact **262** is held by second holding portion **261**. Though the number of second contacts

262 is not particularly limited, for example, nine second contacts are provided. When second rear surface **202** is viewed in the plan view, second contacts **262** are arranged along a line. One of second contacts **262** is, for example, a terminal for electric power supply.

[0075] Second magnetic connection portion **211** is configured, for example, with a pair of magnets. Second magnetic connection portion **211** is located between second main body portion **283** and second front surface **201**. Second magnetic connection portion **211** is arranged inside second device **200**. When second rear surface **202** is viewed in the plan view, second electrical connection portion **260** is located as lying between the pair of magnets. Second magnetic connection portion **211** is magnetically attractable to first magnetic connection portion **111** while it is distant from first magnetic connection portion **111**.

[0076] As shown in FIG. **5**, second device **200** includes a second substrate **231** and a second antenna **232**. Each of second substrate **231** and second antenna **232** is arranged inside second device **200**. More particularly, each of second substrate **231** and second antenna **232** is arranged inside second housing **265**. Second antenna **232** is provided at second substrate **231**.

[0077] The number of second antennas **232** is not particularly limited. The number of second antennas **232** is set, for example, to one or at least two. In electronic device **300** according to the present embodiment, two second antennas **232** are provided. Second antenna **232** may include a second antenna portion **234** and a third antenna portion **236**. Second substrate **231** may include a first substrate portion **233** and a second substrate portion **235**. Second antenna portion **234** is provided at first substrate portion **233**. Third antenna portion **236** is provided at second substrate portion **235**. When second rear surface **202** is viewed in the plan view, second connection portion **220** may be arranged as lying between second antenna portion **234** and third antenna portion **236**.

[0078] As shown in FIG. **5**, when second rear surface **202** is viewed in the plan view, second antenna **232** may be located on an outer side of second connection portion **220** in each of directions toward left and right side surfaces. From another point of view, when second rear surface **202** is viewed in the plan view, second antenna **232** may be arranged at a position not superimposed on second main body portion **283**. For example, resin is employed as a material for second housing **265**. Though the material for second main body portion **283** is not particularly limited, for example, metal may be adopted. Second antenna **232** may be arranged such that second main body portion **283** does not cut off electromagnetic field.

[0079] FIG. **6** is a schematic front view showing a configuration of electronic device **300** in a first connection state. In the first connection state, second rear surface **202** is arranged to cover first display **110**. Second rear surface **202** may be arranged to cover the entire surface or at least a part of first display **110**. Second rear surface **202** is arranged as being superimposed on first front surface **101**. Second front surface **201** faces outward. A user can visually recognize second display **210** arranged at second front surface **201**. In the first connection state, Hall element **130** is arranged as being superimposed on magnet portion **230**. With Hall element **130**, the current connection state may be determined as the first connection state.

[0080] A direction from first left side surface **106** toward first right side surface **105** of first device **100** corresponds to a reference direction B. In the first connection state, when first display **110** is viewed in the plan view, a direction from second left side surface **206** toward second right side surface **205** of second device **200** is a first direction Al. In the first connection state, first device **100** and second device **200** are connected such that the direction from second left side surface **206** toward second right side surface **205** of second device **200** is the same as reference direction B when first display **110** is viewed in the plan view.

[0081] FIG. **7** is a schematic cross-sectional view along the line VII-VII in FIG. **6**. First connection portion **120** is connected to second connection portion **220**. Specifically, first mechanical connection portion **180** is mechanically connected to second mechanical connection portion **280**. First mechanical connection portion **180** includes a first receiving groove **181** and a second receiving groove **182**. Specifically, each of first receiving groove **181** and second receiving groove

182 is provided at a rear side of first main body portion **183**. Second receiving groove **182** is located opposite to first receiving groove **181**. First electrical connection portion **160** is located between first receiving groove **181** and second receiving groove **182**.

[0082] First connection portion **120** is provided with a fourth opening portion **186**. Unlocking button **191** is exposed through fourth opening portion **186**. First device **100** includes a first rotation shaft **192**. Unlocking button **191** is pivotable around first rotation shaft **192**. Fixed hook **281** is arranged in second receiving groove **182** provided at the rear side of first main body portion **183**. Unlocking button **191** may be in contact with fixed hook **281**.

[0083] Second device **200** includes a second rotation shaft **273** and a biasing member **274**. Movable hook **270** is pivotable around second rotation shaft **273**. Movable hook **270** includes a first tab portion **271** and a first support portion **272**. First tab portion **271** is contiguous to first support portion **272**. First support portion **272** is pivotably attached to second rotation shaft **273**. Biasing member **274** biases movable hook **270**. Biasing member **274** is, for example, a spring. Each of second rotation shaft **273** and biasing member **274** is arranged inside second device **200**. A part of movable hook **270** passes through third opening portion **284**.

[0084] As shown in FIGS. **3** and **5**, a width (a second width W**2**) of movable hook **270** is larger than a width (a first width W**1**) of a pair of first protruding portions **184**. Therefore, in the first connection state, movable hook **270** rides on the pair of first protruding portions **184**. In other words, movable hook **270** is not arranged in first receiving groove **181**. As shown in FIG. **7**, in the first connection state, as fixed hook **281** enters second receiving groove **182**, first device **100** and second device **200** are connected. Second device **200** is thus attached to first device **100**. In removal of second device **200** from first device **100**, second device **200** is turned in a first rotation direction R**1**. In this case, unlocking button **191** does not have to be pushed.

[0085] As shown in FIG. **7**, electronic device **300** includes a fourth protruding portion **9**. Fourth protruding portion **9** may be attached to first front surface **101** of first device **100** or second rear surface **202** of second device **200**. In the first connection state, fourth protruding portion **9** is located between first front surface **101** and second rear surface **202**. In the first connection state, first front surface **101** and second rear surface **202** are substantially in parallel to each other. In the first connection state, while second display **210** is in parallel to first display **110**, first device **100** and second device **200** may be connected. With fourth protruding portion **9**, a gap is provided between first front surface **101** and second rear surface **202**. Ninth input portion **121** is distant from second rear surface **202**. Fourth protruding portion **9** can prevent first device **100** and second device **200** from wobbling in the first connection state.

[0086] As shown in FIG. **7**, first electrical connection portion **160** is electrically connected to second electrical connection portion **260**. FIG. **8** is a schematic diagram showing relation between the first connection portion and the second connection portion in the first connection state. The first connection state refers to a state in which second connection portion **220** is connected to first connection portion **120** in first direction Al. In the first connection state, first contact **162** is electrically connected to second contact **262**.

[0087] As shown in FIG. **8**, first contact **162** includes a first terminal **162***a*, a second terminal **162***b*, and a third terminal **162***c*. First electrical connection portion **160** may have terminals disposed in the longitudinal direction of first display **110**. First terminal **162***a* is located closest to the first left side surface. Third terminal **162***c* is located closest to the first right side surface. Second terminal **162***b* is located intermediate between first terminal **162***a* and third terminal **162***c*. A direction from first terminal **162***a* toward third terminal **162***c* corresponds to reference direction B.

[0088] Second contact **262** includes a fourth terminal **262***a*, a fifth terminal **262***b*, and a sixth terminal **262***c*. Second electrical connection portion **260** may have terminals disposed in the longitudinal direction of second display **210**. Fourth terminal **262***a* is located closest to the second left side surface. Sixth terminal **262***c* is located closest to the second right side surface. Fifth terminal **262***b* is located intermediate between fourth terminal **262***a* and sixth terminal **262***c*. In the

first connection state, first terminal **162***a*, second terminal **162***b*, and third terminal **162***c* are electrically connected to fourth terminal **262***a*, fifth terminal **262***b*, and sixth terminal **262***c*, respectively. In the first connection state, second connection portion **220** is arranged with respect to first connection portion **120** such that a direction from fourth terminal **262***a* toward sixth terminal **262***c* is the same as first direction Al. Without being limited to such arrangement of the terminals, first contact **162** and second contact **262** should only be in arrangement relation in which they are connected to each other in the first connection state and the second connection state. For example, a plurality of terminals may be arranged such that, with respect to a terminal arranged at the center, other terminals are in line symmetry.
[0089] FIG. **9** is a schematic cross-sectional view along the line IX-IX in FIG. **6**. As shown in FIG. **9**, first magnetic connection portion **111** is arranged inside first main body portion **183**. First main body portion **183** includes a first inclined surface **187**. First inclined surface **187** is inclined with respect to first front surface **101** of first housing **165**. An angle of inclination of first inclined surface **187** with respect to first front surface **101** is, for example, **10**°. First magnetic connection portion **111** is arranged along first inclined surface **187**. A surface of first magnetic connection portion **111** may substantially be in parallel to first inclined surface **187**.
[0090] Similarly, second magnetic connection portion **211** is arranged inside second main body portion **283**. Second main body portion **283** includes a second inclined surface **287**. Second inclined surface **287** is inclined with respect to second rear surface **202** of second housing **265**. An angle of inclination of second inclined surface **287** with respect to second rear surface **202** is, for example, 10°. Second magnetic connection portion **211** is arranged along second inclined surface **287**. A surface of second magnetic connection portion **211** may substantially be in parallel to second inclined surface **287**. The surface of second magnetic connection portion **211** may substantially be in parallel to the surface of first magnetic connection portion **111**.
[0091] FIG. **10** is a schematic front view showing a configuration of electronic device **300** in the second connection state. In the second connection state, second rear surface **202** is arranged not to cover first display **110**. Therefore, the user can visually recognize each of first display **110** and second display **210**. In the second connection state, a part of first front surface **101** of first device **100** is arranged as being superimposed on a part of second rear surface **202** of second device **200**. In the second connection state, Hall element **130** is arranged as not being superimposed on magnet portion **230**. With Hall element **130**, the current connection state may be determined as the second connection state.
[0092] In the second connection state, when first display **110** is viewed in the plan view, the direction from second left side surface **206** toward second right side surface **205** of second device **200** is a second direction A**2**. In the second connection state, first device **100** and second device **200** are connected such that the direction from second left side surface **206** toward second right side surface **205** of second device **200** is opposite to reference direction B when first display **110** is viewed in the plan view.
[0093] FIG. **11** is a schematic cross-sectional view along the line XI-XI in FIG. **10**. As shown in FIG. **11**, in the second connection state, movable hook **270** is arranged in second receiving groove **182** and fixed hook **281** is arranged in first receiving groove **181**. Movable hook **270** has entered second receiving groove **182**. Fixed hook **281** has entered first receiving groove **181**. First mechanical connection portion **180** is thus mechanically fixed to second mechanical connection portion **280**.
[0094] As shown in FIG. **11**, first electrical connection portion **160** is electrically connected to second electrical connection portion **260**. Specifically, in the second connection state, first contact **162** and second contact **262** are connected. Second rear surface **202** is inclined with respect to first front surface **101**. In the second connection state, first device **100** and second device **200** may be connected while second display **210** is inclined with respect to first display **110**. An angle of inclination of second rear surface **202** with respect to first front surface **101** is, for example, 20°.

[0095] FIG. **12** is a schematic cross-sectional view showing a state in which electronic device **300** in the second connection state is unlocked. As shown in FIG. **12**, unlocking button **191** is pushed into first device **100** along a third direction A**3**. As unlocking button **191** pushes movable hook **270**, movable hook **270** is pushed out of second receiving groove **182**. Consequently, mechanical locking is released. When second device **200** is turned in second rotation direction R**2** in this state, second device **200** is separated from first device **100**.
[0096] FIG. **13** is a schematic diagram showing relation between the first connection portion and the second connection portion in the second connection state. The second connection state refers to a state in which second connection portion **220** is connected to first connection portion **120** in second direction A**2**. Second direction A**2** is opposite to first direction A**1**. In the second connection state, first terminal **162***a*, second terminal **162***b*, and third terminal **162***c* are electrically connected to sixth terminal **262***c*, fifth terminal **262***b*, and fourth terminal **262***a*, respectively. In the second connection state, second connection portion **220** is arranged with respect to first connection portion **120** such that the direction from fourth terminal **262***a* toward sixth terminal **262***c* is the same as second direction A**2**.
[0097] Each of second terminal **162***b* and fifth terminal **262***b* is, for example, a terminal for electric power supply. In each of the first connection state and the second connection state, second terminal **162***b* is electrically connected to fifth terminal **262***b*. Usage of first terminal **162***a* may be the same as usage of third terminal **162***c*. Similarly, usage of fourth terminal **262***a* may be the same as usage of sixth terminal **262***c*. In this case, the same function can be performed in the first connection state and the second connection state. In the first connection state and the second connection state, first contact **162** and second contact **262** may define a plurality of corresponding electrical communication paths. Though usage of first terminal **162***a*, third terminal **162***c*, fourth terminal **262***a*, and sixth terminal **262***c* is not particularly limited, those terminals may be, for example, terminals for grounding, terminals for determination as to whether or not first device **100** and second device **200** are connected, or terminals for resetting the display.
[0098] FIG. **14** is a schematic cross-sectional view showing positional relation between first antenna **132** and second antenna **232** in the first connection state. As shown in FIG. **14**, in the first connection state, first antenna **132** is arranged as being superimposed on second antenna portion **234** of second antenna **232**. In other words, in the first connection state, when first display **110** is viewed in the plan view, first antenna **132** and second antenna **232** are arranged at positions superimposed on each other. Each of first antenna **132** and second antenna **232** is a module in a form of a plate.
[0099] As shown in FIG. **14**, in a cross-section perpendicular to first front surface **101**, first antenna **132** is substantially in parallel to first front surface **101**. First antenna **132** is inclined with respect to first inclined surface **187**. In a cross-section perpendicular to second rear surface **202**, second antenna portion **234** of second antenna **232** is substantially in parallel to second rear surface **202**. Second antenna portion **234** of second antenna **232** is inclined with respect to second inclined surface **287**. In the first connection state, second antenna portion **234** of second antenna **232** is substantially in parallel to first antenna **132**.
[0100] In the first connection state, second inclined surface **287** is in contact with first inclined surface **187**. An angle between first front surface **101** and first inclined surface **187** in the cross-section perpendicular to first front surface **101** is defined as a third angle θ**3**. Third angle θ**3** is, for example, 10°. An angle between second rear surface **202** and second inclined surface **287** in the cross-section perpendicular to second rear surface **202** is defined as a fourth angle θ**4**. Fourth angle θ**4** is, for example, 10°.
[0101] FIG. **15** is a schematic cross-sectional view showing positional relation between first antenna **132** and second antenna **232** in the second connection state. As shown in FIG. **15**, in the second connection state, first antenna portion **132** of first antenna **132** is arranged as being superimposed on third antenna portion **236** of second antenna **232**. In other words, in the second

connection state, when first display **110** is viewed in the plan view, first antenna **132** and second antenna **232** are arranged at positions superimposed on each other. As set forth above, in each of the first connection state and the second connection state, at least a part of first antenna **132** is arranged at a position superimposed on second antenna **232**.

[0102] As shown in FIG. **15**, in the second connection state, third antenna portion **236** of second antenna **232** is inclined with respect to first antenna **132**. In the cross-section perpendicular to each of first front surface **101** and second rear surface **202**, an angle of inclination (a second angle θ**2**) of third antenna portion **236** of second antenna **232** with respect to first antenna **132** is, for example, 20°. In the second connection state, second inclined surface **287** is in contact with first inclined surface **187**.

[0103] Though an example in which first inclined surface **187** and second inclined surface **287** are provided is described with reference to FIGS. **14** and **15**, first inclined surface **187** does not have to be provided in first mechanical connection portion **180** or second inclined surface **287** does not have to be provided in second mechanical connection portion **280**. Specifically, first mechanical connection portion **180** and second mechanical connection portion **280** may be configured to be connected in parallel to each other. In that case, first device **100** and second device **200** may be in parallel to each other, or one of first device **100** or second device **200** may be inclined with respect to the other of first device **100** or second device **200**.

[0104] Though the configuration in which first antenna **132** includes a single antenna portion and second antenna **232** includes two antenna portions is described above, the present disclosure is not limited to the configuration. In electronic device **300** according to another manner, first antenna **132** may include two antenna portions and second antenna **232** may include a single antenna portion. Specifically, second antenna **232** may include the first antenna portion. First antenna **132** may include the second antenna portion and the third antenna portion. In the first connection state, the first antenna portion may be arranged as being superimposed on the second antenna portion. In the second connection state, the first antenna portion may be arranged as being superimposed on the third antenna portion. In electronic device **300** according to yet another manner, first antenna **132** may include two antenna portions and second antenna **232** may include two antenna portions.

[0105] In each of the first connection state and the second connection state, first device **100** and second device **200** can perform first wireless communication through electromagnetic field coupling between first antenna **132** and second antenna **232**. When first antenna **132** and second antenna **232** are electromagnetically coupled to each other, first device **100** and second device **200** may bidirectionally transmit and receive a signal. Second device **200** may unidirectionally receive a signal transmitted from first device **100**. First device **100** may unidirectionally receive a signal transmitted from second device **200**. Though each of first antenna **132** and second antenna **232** is not particularly limited, it is, for example, a 60-GHz millimeter wave module.

Modification of First Embodiment

[0106] FIG. **16** is a schematic cross-sectional view showing a modification of positional

[0107] relation between first antenna **132** and second antenna **232** in the first connection state. As shown in FIG. **16**, in the cross-section perpendicular to first front surface **101**, first antenna **132** may be inclined with respect to first front surface **101**. In the cross-section perpendicular to first front surface **101**, first antenna **132** may be inclined with respect to first inclined surface **187**. Second antenna portion **234** of second antenna **232** may substantially be in parallel to second rear surface **202**. In the first connection state, an angle between first antenna **132** and second antenna **232** in the cross-section perpendicular to each of first front surface **101** and second rear surface **202** is defined as a first angle θ**1**. First angle θ**1** may be smaller than a total of third angle θ**3** and fourth angle θ**4**. Though first angle θ**1** is not particularly limited, it is, for example, not smaller than 2° and not larger than 18°.

[0108] FIG. **17** is a schematic cross-sectional view showing a modification of positional relation between first antenna **132** and second antenna **232** in the second connection state. As shown in

FIG. **17**, in the second connection state, an angle between first antenna **132** and third antenna portion **236** of second antenna **232** in the cross-section perpendicular to each of first front surface **101** and second rear surface **202** is defined as second angle θ**2**. Second angle θ**2** may be smaller than the total of third angle θ**3** and fourth angle θ**4**. Though second angle θ**2** is not particularly limited, it may be smaller than the total of third angle θ**3** and fourth angle θ**4**. For example, second angle θ**2** may be half the total of third angle θ**3** and fourth angle θ**4**. In the present example, second angle θ**2** is 10°. Second angle θ**2** may be not smaller than 2° and not larger than 18°. According to such a configuration, in the first connection state and the second connection state, first antenna **132** and second antenna **232** can be closer to being in parallel. Therefore, for example, in a high frequency band where directivity is strong, a signal can be received with good sensitivity.

[0109] First angle θ**1** may be substantially the same as or different from second angle θ**2**. Though an absolute value of a difference between first angle θ**1** and second angle θ**2** is not particularly limited, it may be not larger than 15° or not larger than 10°.

[0110] Though an example in which first antenna **132** is inclined with respect to first front surface **101** and second antenna **232** is substantially in parallel to second rear surface **202** is described above, electronic device **300** in the present disclosure is not limited as such. Specifically, first antenna **132** may substantially be in parallel to first front surface **101** and second antenna **232** may be inclined with respect to second rear surface **202**, or first antenna **132** may be inclined with respect to first front surface **101** and second antenna **232** may be inclined with respect to second rear surface **202**.

Method of Using Electronic Device

[0111] A method of using electronic device **300** will now be described. Electronic device **300** is, for example, a game console, a smartphone, a tablet personal computer, or the like. First device **100** and second device **200** are usable in both of a case where they are connected to each other and a case where they are separate from each other. As shown in FIGS. **6** and **7**, in the first connection state, first device **100** and second device **200** are connectable. As shown in FIG. **8**, in the first connection state, second connection portion **220** is oriented in first direction Al with respect to first connection portion **120**. As shown in FIGS. **10** and **11**, in the second connection state, first device **100** and second device **200** are connectable. As shown in FIG. **13**, in the second connection state, second connection portion **220** is oriented in second direction A**2** with respect to first connection portion **120**. In the first connection state and the second connection state, first wireless communication is performed through electromagnetic field coupling between first antenna **132** and second antenna **232**.

[0112] In the first connection state and the second connection state, first device **100** may transmit an image signal to second device **200** through first antenna **132** and second antenna **232**. The image signal is, for example, a scalable low voltage signaling (SLVS) signal. For example, in first device **100**, a mobile industry processor interface (MIPI)-display serial interface (DSI) signal is converted to the SLVS signal. The converted SLVS signal is sent from first device **100** to second device **200**. In second device **200**, the SLVS signal is converted to the MIPI-DSI signal. In another manner, the MIPI-DSI signal may directly be transmitted from first device **100** to second device **200** without being converted to the SLVS signal.

[0113] While first device **100** and second device **200** are separate, first device **100** and second device **200** may be able to perform second wireless communication. Second wireless communication while first device **100** and second device **200** are separate may be performed through electromagnetic field coupling between antennas different from each of first antenna **132** and second antenna **232**. Specifically, second wireless communication may be, for example, Wifi® communication. A frequency band of a signal to be used in first wireless communication may be higher than a frequency band of a signal to be used in second wireless communication.

[0114] Second wireless communication is not limited to Wifi® communication. Second wireless communication may be, for example, communication based on Bluetooth® or infrared

communication. Second device **200** may receive a signal from first device **100** and show an image, or second device **200** itself may generate an image without receiving a signal from first device **100**.
[0115] In the first connection state, the user can play a game in a single-screen mode through first wireless communication. In the second connection state, the user can play a game in a dual-screen mode through first wireless communication. While first device **100** and second device **200** are separate, a first user uses first device **100** and a second user uses second device **200**. The first user and the second user can simultaneously play the same game through second wireless communication. In another method of use, one user can operate first device **100** while viewing a screen of second device **200**, with the second device being at rest.
[0116] At least one of first device **100** and second device **200** that configure electronic device **300** may further include a storage (not shown) and/or a processor (not shown). The storage is, for example, a dynamic random access memory (DRAM) or a non-volatile memory. An application program such as a game may be stored in the storage. The processor may be able to perform information processing by reading the application program. First display **110** and second display **210** may show an image, for example, generated as a result of information processing performed by the processor. Electronic device **300** may be an information processing apparatus other than the game console.
Second Embodiment
[0117] A configuration of electronic device **300** according to a second embodiment will now be described. Electronic device **300** according to the second embodiment is different from electronic device **300** according to the first embodiment mainly in configuration of first mechanical connection portion **180** and second mechanical connection portion **280**, and it is otherwise substantially the same as electronic device **300** according to the first embodiment. A configuration different from that of electronic device **300** according to the first embodiment will mainly be described below.
[0118] FIG. **18** is a partial schematic perspective view showing the configuration of electronic device **300** according to the second embodiment. FIG. **19** is a schematic cross-sectional view along the line XIX-XIX in FIG. **18**. As shown in FIGS. **18** and **19**, first mechanical connection portion **180** includes a movable member **140**, a fixed member **150**, and a pair of third biasing portions **154**. Movable member **140** includes a first plate member **141** and a pair of second plate members **142**. When first front surface **101** is viewed in the plan view, first plate member **141** is in a trapezoidal shape. The pair of second plate members **142** is arranged at opposing sides of first plate member **141**. The pair of third biasing portions **154** biases the pair of second plate members **142** in a direction toward first plate member **141**, respectively.
[0119] Fixed member **150** includes an upper surface plate **151** and a partition plate **152**. Upper surface plate **151** is arranged above first magnetic connection portion **111**. Partition plate **152** is attached to upper surface plate **151**. Partition plate **152** extends from upper surface plate **151** toward first rear surface **102**. First magnetic connection portion **111** is arranged between the pair of partition plates **152**.
[0120] Second mechanical connection portion **280** includes a pair of movable hook portions **250**, a pair of rotation shaft portions **253**, a pair of fourth biasing portions **254**, and a lower surface plate **255**. The pair of movable hook portions **250** can pivot around the pair of rotation shaft portions **253**, respectively. Each of the pair of movable hook portions **250** includes a second tab portion **251** and a second support portion **252**. Second tab portion **251** is contiguous to second support portion **252**. Second support portion **252** is pivotably attached to rotation shaft portion **253**. The pair of fourth biasing portions **254** biases the pair of movable hook portions **250**, respectively. Lower surface plate **255** is arranged below second magnetic connection portion **211**.
[0121] As shown in FIG. **19**, when unlocking button **191** is pushed into first device **100** along third direction A**3**, first plate member **141** pushes the pair of second plate members **142** toward opposing sides. Each of the pair of second plate members **142** moves in a fourth direction A**4** perpendicular

to third direction A**3**. The pair of second plate members **142** thus pushes second tab portions **251** of the pair of movable hook portions **250** outward. Consequently, the pair of movable hook portions **250** is unlocked. The pair of unlocked movable hook portions **250** is shown with a dashed line.

[0122] As the user releases unlocking button **191**, each of the pair of second plate members **142** moves to its original position owing to third biasing portion **154**. Each of the pair of movable hook portions **250** moves to its original position owing to fourth biasing portion **254**. Second tab portion **251** of each of the pair of movable hook portions **250** moves to a lower side of upper surface plate **151**. Consequently, second mechanical connection portion **280** is fixed to first mechanical connection portion **180**. The pair of locked movable hook portions **250** is shown with a solid line.

[0123] First electrical connection portion **160** may be provided above or below first display **110**. First mechanical connection portion **180** is provided to sandwich first electrical connection portion **160** from the longitudinal direction of first display **110**. From another point of view, first mechanical connection portion **180** is provided at opposing sides of first electrical connection portion **160**.

[0124] Second electrical connection portion **260** may be provided above or below second display **210**. Second mechanical connection portion **280** is provided to sandwich second electrical connection portion **260** from the longitudinal direction of second display **210**. From another point of view, second mechanical connection portion **280** is provided at opposing sides of second electrical connection portion **260**. First electrical connection portion **160** is electrically connected to second electrical connection portion **260**. First mechanical connection portion **180** is mechanically connected to second mechanical connection portion **280**.

[0125] In the above, unlocking button **191** is pushed in third direction A**3** so that each of the pair of second plate members **142** slides in fourth direction A**4** and each of the pair of movable hook portions **250** is unlocked. A locking mechanism is common to the first connection state and the second connection state. Though the configuration in which a slide mechanism releases locking is described above, an unlocking mechanism in the present disclosure is not limited to the slide mechanism. A link mechanism or a gear mechanism may be employed instead of the slide mechanism.

Third Embodiment

[0126] A configuration of electronic device **300** according to a third embodiment will now be described. Electronic device **300** according to the third embodiment is different from electronic devices **300** according to the first embodiment and the second embodiment mainly in including a first projecting portion **10** and a second projecting portion **20**, and it is otherwise substantially the same as electronic devices **300** according to the first embodiment and the second embodiment. A configuration different from that of electronic devices **300** according to the first embodiment and the second embodiment will mainly be described below.

[0127] FIG. **20** is a schematic front view showing a configuration of first device **100** of electronic device **300** according to the third embodiment. As shown in FIG. **20**, when first display **110** is viewed in the plan view, first housing **165** surrounds first display **110**. First housing **165** includes first projecting portion **10**. First projecting portion **10** extends in an outward direction of first display **110**. First connection portion **120** is provided at a front side of first projecting portion **10**.

[0128] First device **100** of electronic device **300** shown in FIG. **20** is, for example, a game console, a smartphone, a tablet personal computer, a controller that can transmit an input signal to another electronic device, or the like. First device **100** of electronic device **300** does not have to include first display **110**.

[0129] First upper surface **103** includes a part of a sixth input portion **6**, a first left upper surface portion **17**, first projecting portion **10**, a first right upper surface portion **16**, and a part of a second input portion **2**. First projecting portion **10** is formed by connection of a first right inclined portion **12**, a first left inclined portion **13**, a first central portion **11**, a first front surface portion **14** which is a part of first front surface **101**, and a first rear surface portion which is a part of first rear surface

102. First projecting portion **10** is distant from each of first right side surface **105**, first left side surface **106**, and first lower surface **104**. First projecting portion **10** may project only from a portion in first upper surface **103** around a portion intermediate between first front surface **101** and first rear surface **102** and may be distant from each of first front surface **101** and first rear surface **102**.

[0130] First projecting portion **10** includes first right inclined portion **12**, first left inclined portion **13**, first central portion **11**, and first front surface portion **14**. First right inclined portion **12** is located at a first right side surface **105** side of first projecting portion **10**. First left inclined portion **13** is located at a first left side surface **106** side of first projecting portion **10**. First central portion **11** is located between first right inclined portion **12** and first left inclined portion **13**. First central portion **11** is contiguous to each of first right inclined portion **12** and first left inclined portion **13**. First front surface portion **14** is contiguous to each of first right inclined portion **12**, first left inclined portion **13**, and first central portion **11**. First front surface portion **14** defines a part of first front surface **101**.

[0131] As shown in FIG. **20**, when first display **110** is viewed in the plan view, first central portion **11** is substantially in parallel to first lower surface **104**. Each of first right inclined portion **12** and first left inclined portion **13** is inclined with respect to first central portion **11**. A boundary portion between first central portion **11** and first right inclined portion **12** may be curved as convexly projecting outward. Similarly, a boundary portion between first central portion **11** and first left inclined portion **13** may be curved as convexly projecting outward.

[0132] First upper surface **103** includes first right upper surface portion **16** and first left upper surface portion **17**. First right upper surface portion **16** is located between first right inclined portion **12** and first right side surface **105**. First right upper surface portion **16** is contiguous to each of first right inclined portion **12** and first right side surface **105**. A boundary portion between first right upper surface portion **16** and first right side surface **105** may be curved as convexly projecting outward. First right inclined portion **12** is a portion inclined from first central portion **11** toward first right upper surface portion **16**.

[0133] Similarly, first left upper surface portion **17** is located between first left inclined portion **13** and first left side surface **106**. First left upper surface portion **17** is contiguous to each of first left inclined portion **13** and first left side surface **106**. A boundary portion between first left upper surface portion **17** and first left side surface **106** may be curved as convexly projecting outward. First left inclined portion **13** is a portion inclined from first central portion **11** toward first left upper surface portion **17**.

[0134] First device **100** further includes a first input portion **1**, second input portion **2**, a fifth input portion **5**, and sixth input portion **6**. First input portion **1** is arranged at first projecting portion **10**. Specifically, first input portion **1** is provided at a portion of first projecting portion **10** at the first right side surface **105***a* side. When first display **110** is viewed in the plan view, first input portion **1** is provided closer to first right side surface **105** relative to a center of first upper surface **103**. First input portion **1** is provided at first right inclined portion **12**. First input portion **1** may be provided at a boundary portion between first central portion **11** and first right inclined portion **12**.

[0135] Second input portion **2** is provided at first upper surface **103** other than first projecting portion **10**. Specifically, second input portion **2** is provided at first right upper surface portion **16**. Second input portion **2** may be provided at a boundary portion between first right upper surface portion **16** and first right side surface **105**. Second input portion **2** is provided closer to first right side surface **105** relative to first input portion **1**. In a direction from first lower surface **104** toward first upper surface **103**, a lowermost portion of first input portion **1** may be located above an uppermost portion of second input portion **2**.

[0136] Fifth input portion **5** is arranged at first projecting portion **10**. Specifically, fifth input portion **5** is provided at a portion of first projecting portion **10** at the first left side surface **106** side. When first display **110** is viewed in the plan view, fifth input portion **5** is provided closer to first left side surface **106** relative to the center of first upper surface **103**. Fifth input portion **5** is provided at first left inclined portion **13**. Fifth input portion **5** may be provided at a boundary portion between first central portion **11** and first left inclined portion **13**.

[0137] Sixth input portion **6** is provided at first upper surface **103** other than first projecting portion **10**. Specifically, sixth input portion **6** is provided at first left upper surface portion **17**. Sixth input portion **6** may be provided at a boundary portion between first left upper surface portion **17** and first left side surface **106**. Sixth input portion **6** is provided closer to first left side surface **106** relative to fifth input portion **5**. In the direction from first lower surface **104** toward first upper surface **103**, a lowermost portion of fifth input portion **5** may be located above an uppermost portion of sixth input portion **6**.

[0138] As shown in FIG. **20**, when first display **110** is viewed in the plan view, first input portion **1** may be located closer to first right side surface **105** relative to first connection portion **120**. Similarly, when first display **110** is viewed in the plan view, fifth input portion **5** may be located closer to first left side surface **106** relative to first connection portion **120**.

[0139] FIG. **21** is a schematic cross-sectional view showing a configuration of fifth input portion **5** and sixth input portion **6**. The cross-section shown in FIG. **21** is in parallel to first front surface **101**. First device **100** includes a support member **40**, a shaft receiving portion **41**, an elastic body **44**, a hole **42**, and a detection portion **43**. Shaft receiving portion **41** and hole **42** are recesses provided in support member **40**. Elastic body **44** is, for example, a coil spring. Elastic body **44** is arranged in hole **42**. Detection portion **43** is fixed to support member **40**. Support member **40**, shaft receiving portion **41**, elastic body **44**, hole **42**, and detection portion **43** are arranged inside first housing **165**.

[0140] Fifth input portion **5** includes a pressed portion **53**, a movable shaft **51**, a stopper portion **52**, a second protruding portion **54**, and a third protruding portion **55**. Pressed portion **53** is a portion to be pushed by the user. Movable shaft **51** is arranged in shaft receiving portion **41**. Movable shaft **51** is contiguous to pressed portion **53**. Movable shaft **51** is in contact with an inner wall of first housing **165**. Stopper portion **52** is contiguous to pressed portion **53**. Stopper portion **52** is located opposite to movable shaft **51** with respect to pressed portion **53**. Stopper portion **52** is in contact with the inner wall of first housing **165**.

[0141] Second protruding portion **54** is contiguous to pressed portion **53**. Second protruding portion **54** is arranged as being in contact with detection portion **43**. Third protruding portion **55** is contiguous to pressed portion **53**. Third protruding portion **55** is attached to elastic body **44**. First projecting portion **10** of first housing **165** is provided with a fifth opening portion **70**. Pressed portion **53** is exposed to the outside of first housing **165** through fifth opening portion **70**.

[0142] When the user pushes pressed portion **53** with his/her finger, second protruding portion **54** moves to the inside of first housing **165**. As second protruding portion **54** applies a pressure to detection portion **43**, input by the user is detected. As the user's finger moves away from pressed portion **53**, fifth input portion **5** returns to an original position owing to repulsive force of elastic body **44**. Each of movable shaft **51** and stopper portion **52** is in contact with the inner wall of first housing **165**.

[0143] Though the configuration of fifth input portion **5** is described above, other input portions (specifically, first input portion **1**, second input portion **2**, a third input portion **3**, a fourth input portion **4**, sixth input portion **6**, a seventh input portion **7**, and an eighth input portion **8**) are substantially the same in configuration to fifth input portion **5**. Therefore, description of the configuration of other input portions will not be provided.

[0144] FIG. **22** is a schematic front view showing a configuration of second device **200** of electronic device **300** according to the third embodiment. As shown in FIG. **22**, when second display **210** is viewed in the plan view, second housing **265** surrounds second display **210**. Second housing **265** includes a second projecting portion **20**. Second projecting portion **20** extends in the outward direction of second display **210**. Second connection portion **220** may be provided at a rear side of second projecting portion **20**.

[0145] Second projecting portion **20** is provided at second upper surface **203**. Second projecting portion **20** may define a part of second front surface **201**. Similarly, second projecting portion **20** may define a part of second rear surface **202**. Second projecting portion **20** is distant from each of second right side surface **205**, second left side surface **206**, and second lower surface **204**.

[0146] Second projecting portion **20** includes a second right inclined portion **22**, a second left inclined portion **23**, a second central portion **21**, and a second front surface portion **24**. Second right inclined portion **22** is located at a second right side surface **205** side of second projecting portion **20**. Second left inclined portion **23** is located at a second left side surface **206** side of second projecting portion **20**. Second central portion **21** is located between second right inclined portion **22** and second left inclined portion **23**. Second central portion **21** is contiguous to each of second right inclined portion **22** and second left inclined portion **23**. Second front surface portion **24** is contiguous to each of second right inclined portion **22**, second left inclined portion **23**, and second central portion **21**. Second front surface portion **24** defines a part of second front surface **201**.

[0147] As shown in FIG. **22**, when second display **210** is viewed in the plan view, second central portion **21** is substantially in parallel to second lower surface **204**. Each of second right inclined portion **22** and second left inclined portion **23** is inclined with respect to second central portion **21**. A boundary portion between second central portion **21** and second right inclined portion **22** may be curved as convexly projecting outward. Similarly, a boundary portion between second central portion **21** and second left inclined portion **23** may be curved as convexly projecting outward.

[0148] Second upper surface **203** includes a second right upper surface portion **26** and a second left upper surface portion **27**. Second right upper surface portion **26** is located between second right inclined portion **22** and second right side surface **205**. Second right upper surface portion **26** is contiguous to each of second right inclined portion **22** and second right side surface **205**. A boundary portion between second right upper surface portion **26** and second right side surface **205** may be curved as convexly projecting outward.

[0149] Similarly, second left upper surface portion **27** is located between second left inclined portion **23** and second left side surface **206**. Second left upper surface portion **27** is contiguous to each of second left inclined portion **23** and second left side surface **206**. A boundary portion between second left upper surface portion **27** and second left side surface **206** may be curved as convexly projecting outward.

[0150] Second device **200** further includes third input portion **3**, fourth input portion **4**, seventh input portion **7**, and eighth input portion **8**. Third input portion **3** is arranged at second projecting portion **20**. Specifically, third input portion **3** is provided at a portion of second projecting portion **20** at the second right side surface **205** side. When second display **210** is viewed in the plan view, third input portion **3** is provided closer to second right side surface **205** relative to a center of second upper surface **203**. Third input portion **3** is provided at second right inclined portion **22**. Third input portion **3** may be provided at a boundary portion between second central portion **21** and second right inclined portion **22**.

[0151] Fourth input portion **4** is provided at second upper surface **203** other than second projecting portion **20**. Specifically, fourth input portion **4** is provided at second right upper surface portion **26**. Fourth input portion **4** may be provided at a boundary portion between second right upper surface portion **26** and second right side surface **205**. Fourth input portion **4** is provided closer to second right side surface **205** relative to third input portion **3**. In a direction from second lower surface **204** toward second upper surface **203**, a lowermost portion of third input portion **3** may be located above an uppermost portion of fourth input portion **4**.

[0152] Seventh input portion **7** is arranged at second projecting portion **20**. Specifically, seventh input portion **7** is provided at a portion of second projecting portion **20** at the second left side surface **206** side. When second display **210** is viewed in the plan view, seventh input portion **7** is provided closer to second left side surface **206** relative to the center of second upper surface **203**.

Seventh input portion **7** is provided at second left inclined portion **23**. Seventh input portion **7** may be provided at a boundary portion between second central portion **21** and second left inclined portion **23**.

[0153] Eighth input portion **8** is provided at second upper surface **203** other than second projecting portion **20**. Specifically, eighth input portion **8** is provided at second left upper surface portion **27**. Eighth input portion **8** may be provided at a boundary portion between second left upper surface portion **27** and second left side surface **206**. Eighth input portion **8** is provided closer to second left side surface **206** relative to seventh input portion **7**. In the direction from second lower surface **204** toward second upper surface **203**, a lowermost portion of seventh input portion **7** may be located above an uppermost portion of eighth input portion **8**.

[0154] As shown in FIG. **22**, a front surface (second front surface portion **24**) of second projecting portion **20** may be provided with a functional component **29**. Functional component **29** may be, for example, a camera lens or an input portion. Second front surface **201** other than second projecting portion **20** may be provided with an operation portion (not shown). For example, second front surface **201** other than second projecting portion **20** may be provided with a physical stick. According to such a configuration, a space at the front surface (second front surface portion **24**) of second projecting portion **20** can effectively be made use of.

[0155] FIG. **23** is a schematic cross-sectional view along the line XXIII-XXIII in FIG. **22**. As shown in FIG. **23**, second display **210** includes a touch panel **32** and a touch panel cover **31**. Touch panel **32** is a panel on which a touch operation can be performed. Touch panel **32** performs a function to detect touch. Touch panel cover **31** covers touch panel **32**. Touch panel cover **31** can receive an input operation to touch panel **32**. Touch panel cover **31** itself does not perform a function to detect touch. When the user touches touch panel cover **31**, the input operation to touch panel **32** therethrough can be performed.

[0156] Second display **210** includes a first tactile feel presentation portion **81** and a second tactile feel presentation portion **82**. First tactile feel presentation portion **81** is implemented, for example, by touch panel cover **31**. Touch panel cover **31** is provided with an opening **30**. Second tactile feel presentation portion **82** is, for example, a portion where touch panel **32** is exposed through opening **30**. Second tactile feel presentation portion **82** is a part of touch panel **32**. Second tactile feel presentation portion **82** may be used as a virtual button or a virtual stick.

[0157] Second tactile feel presentation portion **82** may be a part of touch panel cover **31**. In this case, second tactile feel presentation portion **82** may be different from first tactile feel presentation portion **81** in sense of touch. Second tactile feel presentation portion **82** may be a portion of a surface of touch panel cover **31** that is roughened. In this case, first tactile feel presentation portion **81** is a portion of the surface of touch panel cover **31** that is not roughened. Second tactile feel presentation portion **82** may be a portion where vibration is generated as the user touches the same. In this case, first tactile feel presentation portion **81** is a portion where vibration is not generated even when the user touches the same.

[0158] FIG. **24** is a schematic perspective view showing a configuration of electronic device **300** according to the third embodiment in the first connection state. As shown in FIG. **24**, in the first connection state, second rear surface **202** is arranged to cover first display **110**. First projecting portion **10** is arranged as being superimposed on second projecting portion **20**. Specifically, the front surface of first projecting portion **10** is superimposed on the rear surface of second projecting portion **20**. First connection portion **120** provided at the front surface of first projecting portion **10** is connected, in first direction Al, to second connection portion **220** provided at the rear surface of second projecting portion **20**.

[0159] In the first connection state, in the direction perpendicular to first front surface **101**, first input portion **1** and third input portion **3** are arranged as being aligned and second input portion **2** and fourth input portion **4** are arranged as being aligned. From another point of view, along the direction from first front surface **101** toward second rear surface **202**, first input portion **1** and third input portion **3** are arranged as being aligned and second input portion **2** and fourth input portion **4** are arranged as being aligned.

[0160] In the first connection state, in the direction perpendicular to first front surface **101**, fifth input portion **5** and seventh input portion **7** are arranged as being aligned and sixth input portion **6** and eighth input portion **8** are arranged as being aligned. From another point of view, along the direction from first front surface **101** toward second rear surface **202**, fifth input portion **5** and seventh input portion **7** are arranged as being aligned and sixth input portion **6** and eighth input portion **8** are arranged as being aligned.

[0161] FIG. **25** is a schematic perspective view showing a configuration of electronic device **300** according to the third embodiment in the second connection state. In the second connection state, second rear surface **202** is arranged not to cover first display **110**. First projecting portion **10** is arranged as being superimposed on second projecting portion **20**. Specifically, the front surface of first projecting portion **10** is superimposed on the rear surface of second projecting portion **20**. First connection portion **120** provided at the front surface of first projecting portion **10** is connected, in second direction A**2**, to second connection portion **220** provided at the rear surface of second projecting portion **20**.

[0162] In the second connection state, first device **100** and second device **200** are connected such that first projecting portion **10** is superimposed on second projecting portion **20**, and hence first input portion **1**, second input portion **2**, fifth input portion **5**, and sixth input portion **6** are structured to be distant from second device **200**. Therefore, when each of first input portion **1**, second input portion **2**, fifth input portion **5**, and sixth input portion **6** is operated, the user's finger is less likely to interfere with third input portion **3**, fourth input portion **4**, seventh input portion **7**, and eighth input portion **8** or second projecting portion **20** of second device **200**. Operability of electronic device **300** is thus improved.

[0163] Electronic device **300** according to a modification of the third embodiment does not have to include first input portion **1** to eighth input portion **8**. Specifically, first device **100** does not have to include each of first input portion **1**, second input portion **2**, fifth input portion **5**, and sixth input portion **6**, and second device **200** does not have to include each of third input portion **3**, fourth input portion **4**, seventh input portion **7**, and eighth input portion **8**.

[0164] Though the configuration in which second rear surface **202** covers first display **110** in the first connection state and second rear surface **202** does not cover first display **110** in the second connection state is disclosed, in the first connection state in which second connection portion **220** is connected to first connection portion **120** in first direction Al, second rear surface **202** may be arranged to cover input portion **121**, and in the second connection state in which second connection portion **220** is connected to first connection portion **120** in second direction A**2**, second rear surface **202** may be arranged not to cover input portion **121**.

[0165] First input portion **1** may be configured to extend to first right upper surface portion **16**. Third input portion **3** may be configured to extend to second right upper surface portion **26**. Fifth input portion **5** may be configured to extend to first left upper surface portion **17**. Seventh input portion **7** may be configured to extend to second left upper surface portion **27**.

[0166] First input portion **1** and second input portion **2** may integrally be formed. A function as a button may be provided to a portion of each of first input portion **1** and second input portion **2**. Third input portion **3** and fourth input portion **4** may integrally be formed. A function as a button may be provided to a portion of each of third input portion **3** and fourth input portion **4**. Fifth input portion **5** and sixth input portion **6** may integrally be formed. A function as a button may be provided to a portion of each of fifth input portion **5** and sixth input portion **6**. Seventh input portion **7** and eighth input portion **8** may integrally be formed. A function as a button may be provided to a portion of each of seventh input portion **7** and eighth input portion **8**.

[0167] Although the present disclosure has been described and illustrated in detail, it is clearly understood that the same is by way of illustration and example only and is not to be taken by way of limitation, the scope of the present disclosure being interpreted by the terms of the appended claims.

Claims

1. An electronic device comprising: a first device; and a second device, wherein the first device and the second device are removably attachable to each other, the first device comprises a first front surface, and a first display and a first connection portion arranged at the first front surface, the second device comprises a second front surface, a second display arranged at the second front surface, a second rear surface located opposite to the second front surface, and a second connection portion arranged at the second rear surface, the second connection portion is connectable to the first connection portion in a first direction and a second direction opposite to the first direction, in a first connection state in which the second connection portion is connected to the first connection portion in the first direction, the second rear surface is arranged to cover the first display, and in a second connection state in which the second connection portion is connected to the first connection portion in the second direction, the second rear surface is arranged not to cover the first display.
2. The electronic device according to claim 1, wherein the first device comprises a first substrate arranged inside the first device, and one first antenna or at least two first antennas provided at the first substrate, the second device comprises a second substrate arranged inside the second device, and one second antenna or at least two second antennas provided at the second substrate, in each of the first connection state and the second connection state, when the first display is viewed in a plan view, the first antenna and the second antenna are arranged at positions superimposed on each other, and the first device and the second device are configured to perform first wireless communication through electromagnetic field coupling between the first antenna and the second antenna.
3. The electronic device according to claim 2, wherein in each of the first connection state and the second connection state, at least a part of the first antenna is arranged at a position superimposed on the second antenna.
4. The electronic device according to claim 3, wherein one of the first antenna or the second antenna comprises a first antenna portion, the other of the first antenna or the second antenna comprises a second antenna portion and a third antenna portion, in the first connection state, the first antenna portion is arranged as being superimposed on the second antenna portion, and in the second connection state, the first antenna portion is arranged as being superimposed on the third antenna portion.
5. The electronic device according to claim 4, wherein the first antenna comprises the first antenna portion, the second antenna comprises the second antenna portion and the third antenna portion, when the first front surface is viewed in the plan view, the first antenna portion is arranged at a position adjacent to the first connection portion, and when the second rear surface is viewed in the plan view, the second connection portion is arranged as lying between the second antenna portion and the third antenna portion.
6. The electronic device according to claim 1, wherein the first device comprises a first substrate arranged inside the first device, and one first antenna or at least two first antennas provided at the first substrate, the second device comprises a second substrate arranged inside the second device, and one second antenna or at least two second antennas provided at the second substrate, one of the first antenna or the second antenna comprises a first antenna portion, the other of the first antenna or the second antenna comprises a second antenna portion and a third antenna portion, in the first connection state, the second antenna portion is arranged as being superimposed on the first antenna portion and the third antenna portion is arranged as not being superimposed on the first antenna, and in the second connection state, the third antenna portion is arranged as being superimposed on the first antenna portion and the second antenna portion is arranged as not being superimposed on

the first antenna.

7. The electronic device according to claim 6, wherein when the first front surface is viewed in a plan view, the first antenna portion is arranged at a position adjacent to the first connection portion, and when the second rear surface is viewed in the plan view, the second connection portion is arranged as lying between the second antenna portion and the third antenna portion.

8. The electronic device according to claim 2, wherein in the second connection state, the first device and the second device are connected with the second display being inclined with respect to the first display.

9. The electronic device according to claim 2, wherein in the first connection state, the first device and the second device are connected with the second display being in parallel to the first display.

10. The electronic device according to claim 8, wherein the first connection portion comprises a first inclined surface inclined with respect to the first front surface, and the second connection portion comprises a second inclined surface inclined with respect to the second rear surface.

11. The electronic device according to claim 10, wherein when an angle between the first antenna and the second antenna in a cross-section perpendicular to each of the first front surface and the second rear surface in the first connection state is defined as a first angle, an angle between the first antenna and the second antenna in the cross-section perpendicular to each of the first front surface and the second rear surface in the second connection state is defined as a second angle, an angle between the first front surface and the first inclined surface in the cross-section perpendicular to the first front surface is defined as a third angle, and an angle between the second rear surface and the second inclined surface in the cross-section perpendicular to the second rear surface is defined as a fourth angle, each of the first angle and the second angle is smaller than a total of the third angle and the fourth angle.

12. The electronic device according to claim 2, wherein when the first antenna and the second antenna are electromagnetically coupled to each other, the first device and the second device transmit and receive a signal unidirectionally or bidirectionally, the first connection portion comprises a first contact for electric power supply, the second connection portion comprises a second contact for electric power supply, and in each of the first connection state and the second connection state, the first contact and the second contact are connected.

13. The electronic device according to claim 2, wherein the first device and the second device are configured to perform second wireless communication while the first device and the second device are separate.

14. The electronic device according to claim 13, wherein the first wireless communication in the first connection state and the second connection state is performed through electromagnetic field coupling between the first antenna and the second antenna, the second wireless communication while the first device and the second device are separate is performed through electromagnetic field coupling between antennas different from each of the first antenna and the second antenna, and a frequency band of a signal to be used in the first wireless communication is higher than a frequency band of a signal to be used in the second wireless communication.

15. The electronic device according to claim 1, wherein the first connection portion comprises a first receiving groove, and a second receiving groove located opposite to the first receiving groove, the second connection portion comprises a movable hook, and a fixed hook located opposite to the movable hook, in the first connection state, the movable hook is not arranged in the first receiving groove and the fixed hook is arranged in the second receiving groove, and in the second connection state, the movable hook is arranged in the second receiving groove and the fixed hook is arranged in the first receiving groove.

16. The electronic device according to claim 1, wherein the first connection portion comprises a first electrical connection portion and a first mechanical connection portion, the first electrical connection portion is provided above or below the first display, the first mechanical connection portion is provided to sandwich the first electrical connection portion from a longitudinal direction of the first display, the second connection portion comprises a second electrical connection portion and a second mechanical connection portion, the second electrical connection portion is provided above or below the second display, the second mechanical connection portion is provided to sandwich the second electrical connection portion from a longitudinal direction of the second display, the first electrical connection portion is to electrically be connected to the second electrical connection portion, and the first mechanical connection portion is to mechanically be connected to the second mechanical connection portion.

17. The electronic device according to claim 16, wherein the first electrical connection portion has terminals disposed in the longitudinal direction of the first display, and the second electrical connection portion has a terminal disposed in the longitudinal direction of the second display.

18. The electronic device according to claim 1, wherein the first device comprises an input portion configured to be operated as a stick or a button, and the second display comprises a touch panel on which a touch input can be provided.

19. The electronic device according to claim 1, wherein the first device comprises a first housing that surrounds the first display and comprises a first projecting portion that extends in an outward direction of the first display, the second device comprises a second housing that surrounds the second display and comprises a second projecting portion that extends in an outward direction of the second display, the first connection portion is provided at a front side of the first projecting portion, and the second connection portion is provided at a rear side of the second projecting portion.

20. The electronic device according to claim 19, wherein the first device comprises a first rear surface located opposite to the first front surface, a first upper surface contiguous to the first front surface and the first rear surface, a first lower surface located opposite to the first upper surface, a first right side surface contiguous to each of the first front surface, the first rear surface, the first upper surface, and the first lower surface, and a first left side surface located opposite to the first right side surface, the second device comprises a second upper surface contiguous to the second front surface and the second rear surface, a second lower surface located opposite to the second upper surface, a second right side surface contiguous to each of the second front surface, the second rear surface, the second upper surface, and the second lower surface, and a second left side surface located opposite to the second right side surface, the first projecting portion is provided at the first upper surface, the second projecting portion is provided at the second upper surface, the first device comprises a first input portion provided at a portion of the first projecting portion at a side of the first right side surface, and a second input portion provided at the first upper surface other than the first projecting portion, and the second device comprises a third input portion provided at a portion of the second projecting portion at a side of the second right side surface, and a fourth input portion provided at the second upper surface other than the second projecting portion.

21. The electronic device according to claim 20, wherein in the first connection state, in a direction perpendicular to the first front surface, the first input portion and the third input portion are arranged as being aligned and the second input portion and the fourth input portion are arranged as being aligned.

22. The electronic device according to claim 19, wherein at a surface of the second projecting portion, an input portion or a camera lens is provided.

23. The electronic device according to claim 1, wherein the second display comprises a touch panel on which a touch operation can be performed, and a touch panel cover on which an input operation to the touch panel can be performed, the touch panel cover covering the touch panel, and the second display comprises a first tactile feel presentation portion configured with the touch panel cover, and a second tactile feel presentation portion where the touch panel is exposed in an opening provided in the touch panel cover.